PROGRAM OVERVIEW

# Cap Collective

## Turning Bottle Caps into Community Impact

*"Join the Bridge. Create the Impact."*

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# Table of Contents


**1. Cap Collective — Program Overview**

- What Is Cap Collective?
- The Problem in One Line
- The Solution: A 5-Step Micro-Factory in Your School
- Why Caps?
- What Students Make
- Impact You Can Measure
- Three Ways to Start (Any Budget)
- Learn While You Build
- Safety Is Non-Negotiable
- The Complete Implementation Kit
- How to Get Started
- Our Guiding Idea

# Cap Collective — Program Overview

> *"The distance between an idea and its impact is the bridge we build today."*

## What Is Cap Collective?

Cap Collective is a **student-led circular recycling program** that captures discarded plastic bottle caps and transforms them into useful products — while funding the school garden. Students run every part of it: collection, cleaning, shredding, melting, product design, sales, and impact reporting.

Bottle caps are a **recycling blind spot**. Mechanical recycling centers process large items like bottles, but caps are too small — they fall through sorting screens or jam high-speed equipment. The result: a perfectly good, high-grade plastic is thrown away, every day, in every school.

Cap Collective captures that resource **at the source** and turns it into value.

## The Problem in One Line

Only about **3% of bottle caps** are recycled globally. The rest end up in landfill or the ocean — where they look like food to sea turtles and fish.

## The Solution: A 5-Step Micro-Factory in Your School

```
1. COLLECT   →  HDPE (#2) & PP (#5) caps dropped in campus boxes
2. CLEAN     →  Remove liners · wash · air-dry 24 hours
3. SHRED     →  Uniform 5–8 mm flakes
4. MELT      →  Heat press: HDPE 140°C · PP 170°C  (never above 200°C)
5. REFINE    →  Laser-cut · injection-mold · press into finished goods
```

**Closed loop:** When a product breaks, it goes right back to Step 3. 100% recyclable, forever.

## Why Caps?

| Property | Why It Matters |
|---|---|
| **HDPE (#2) & PP (#5)** | Melt predictably; safe when heated carefully |
| **High density & color** | Strong products with beautiful marbled finishes |
| **Abundant** | Every school produces thousands per term |
| **Food-safe grade** | Suitable for useful everyday items |
| **Ignored by industry** | An untapped resource, not a nuisance |

No PVC (#3) is ever used — it releases toxic gas when heated.

## What Students Make

Coasters · keychains · phone stands · desk organizers · plant pots · art tiles · and more.

Each product uses a known number of caps (e.g., a keychain ≈ 4 caps, a coaster ≈ 15 caps), so students can see — and calculate — exactly how waste becomes value.

## Impact You Can Measure

The program is built around real, trackable numbers. Targets **per semester**:

| Metric | Target |
|---|---|
| Caps collected | 10,000+ |
| Plastic diverted | 35+ kg |
| $CO_2e$ avoided | 65+ kg |
| Product revenue | $500+ |
| Student participation | 80% |

A live dashboard counts caps automatically (via an optional sensor) and shows the impact in real time.

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## Three Ways to Start (Any Budget)

| Tier | Cost | Best For |
|---|---|---|
| **Core** | $500 – $1,000 | Primary schools, clubs, pilots |
| **FabLab** | $2,500 – $5,000 | Middle/high school makerspaces |
| **Center** | $10,000+ | Technical schools, district hubs |

Start small. A pilot with boxes, a wash station, and a heat press proves the concept. Expand from there.

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## Learn While You Build

Cap Collective is a **full curriculum platform**, not just a recycling bin. It spans four grade bands:

- **K-2 — Cap Detectives:** observation, sorting, storytelling
- **3-5 — Cap Engineers:** measurement, building, basic design
- **6-8 — Cap Innovators:** data, sensors, entrepreneurship
- **9-12 — Cap Systems Designers:** engineering, life-cycle analysis, policy, scale

Every lesson links to standards (NGSS, Common Core, IB) and comes with printable worksheets.

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## Safety Is Non-Negotiable

- **No certification = no machine access.**
- Maximum temperature: **200°C** — never exceeded.
- Ventilation mandatory for all heating.
- PPE required per station.
- Every machine has a Standard Operating Procedure and a student certification card.

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## The Complete Implementation Kit

This program ships with **29 documents** covering the entire journey — from convincing your principal to handing the program off to the next year's students:

**Plan & Pitch** · Readiness audit · Launch deck · Speaker notes · School buy-in & approvals **Fund & Buy** · Procurement, payment & finance · Grant & fundraising templates **Run It** · Teacher guide · Command Center sheets · Safety SOPs & risk assessment **Teach It** · Lesson plans · Student worksheets · Curriculum mapping · Certification & training **Launch It** · Launch-week run-of-show · Communications kit · Printable resources **Sell It** · Eco-Store operations · Product development & QC **Grow It** · Engagement & gamification · Accessibility & inclusion · Impact reporting **Scale It** · District replication · Continuity & handover · Data science & AI extensions

Plus: importable spreadsheets, parametric CAD starters, and sensor code.

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## How to Get Started

1. **Read** the Implementation Readiness Audit — an honest picture of what it takes.

2. **Pitch** using the Launch Deck to your school leaders.
3. **Approve & fund** with the buy-in and finance playbooks.
4. **Train** staff and students with the certification pack.
5. **Launch** with the run-of-show and communications kit.
6. **Measure & report** with the dashboard and reporting templates.
7. **Scale & sustain** using the district and continuity guides.

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## Our Guiding Idea

> *Join the Bridge. Create the Impact.*

Cap Collective proves that sustainability isn't something students learn about — it's something students **build**.

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*Cap Collective — Empowering Sustainability through Environmental Stewardship and Economic Innovation.*